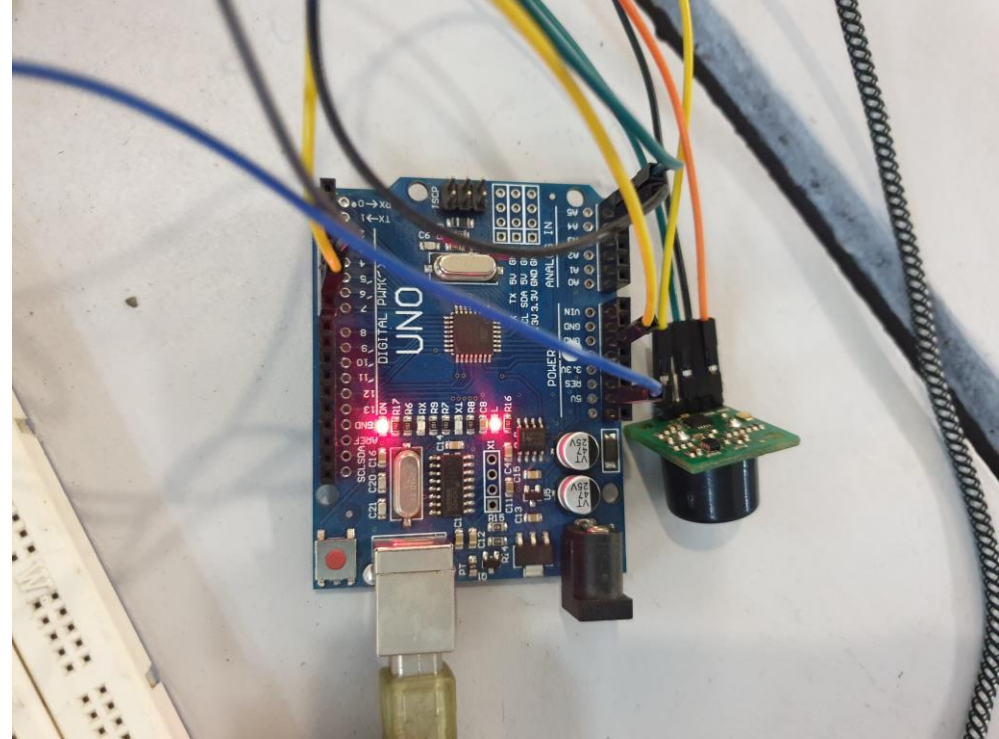
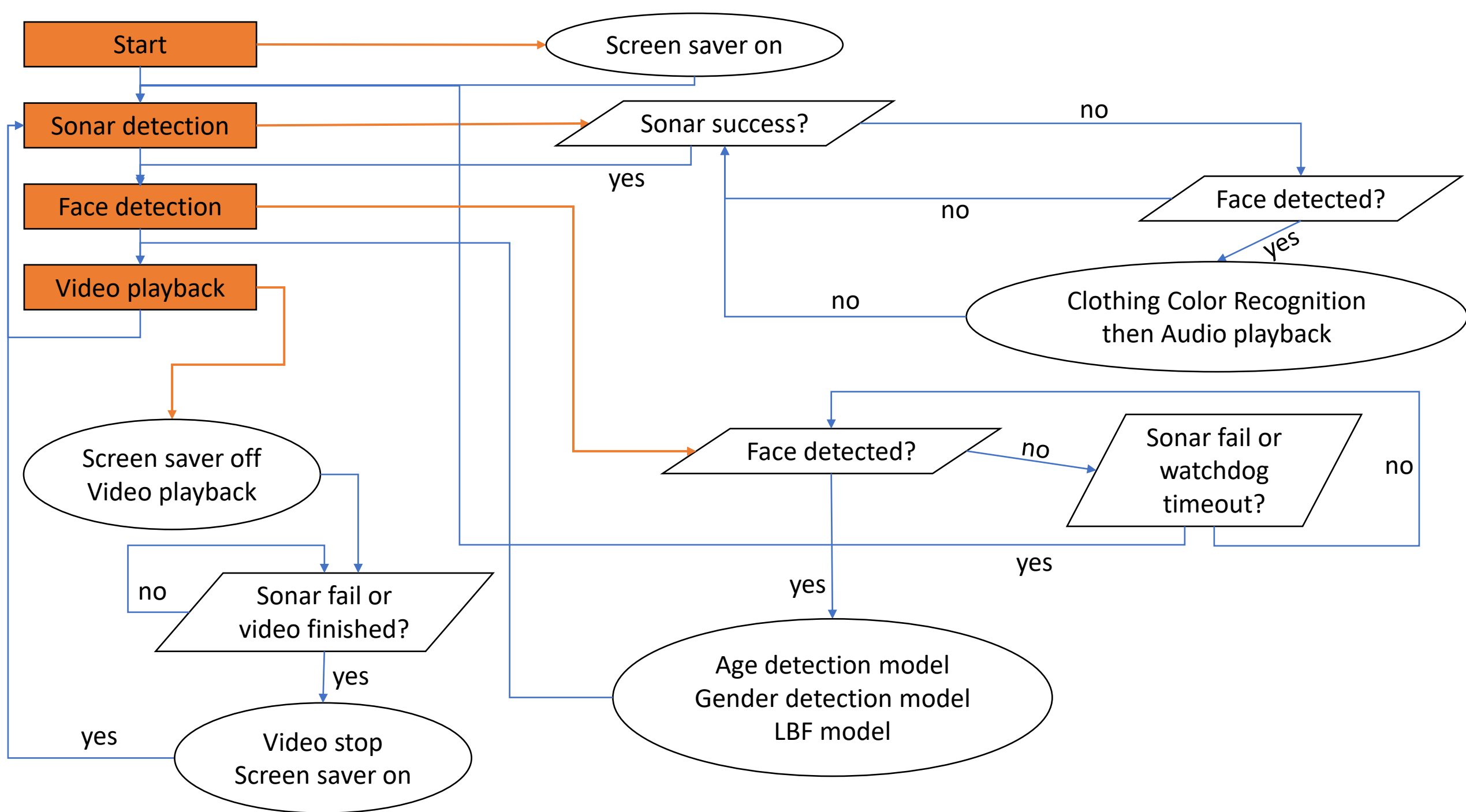


Salam robot

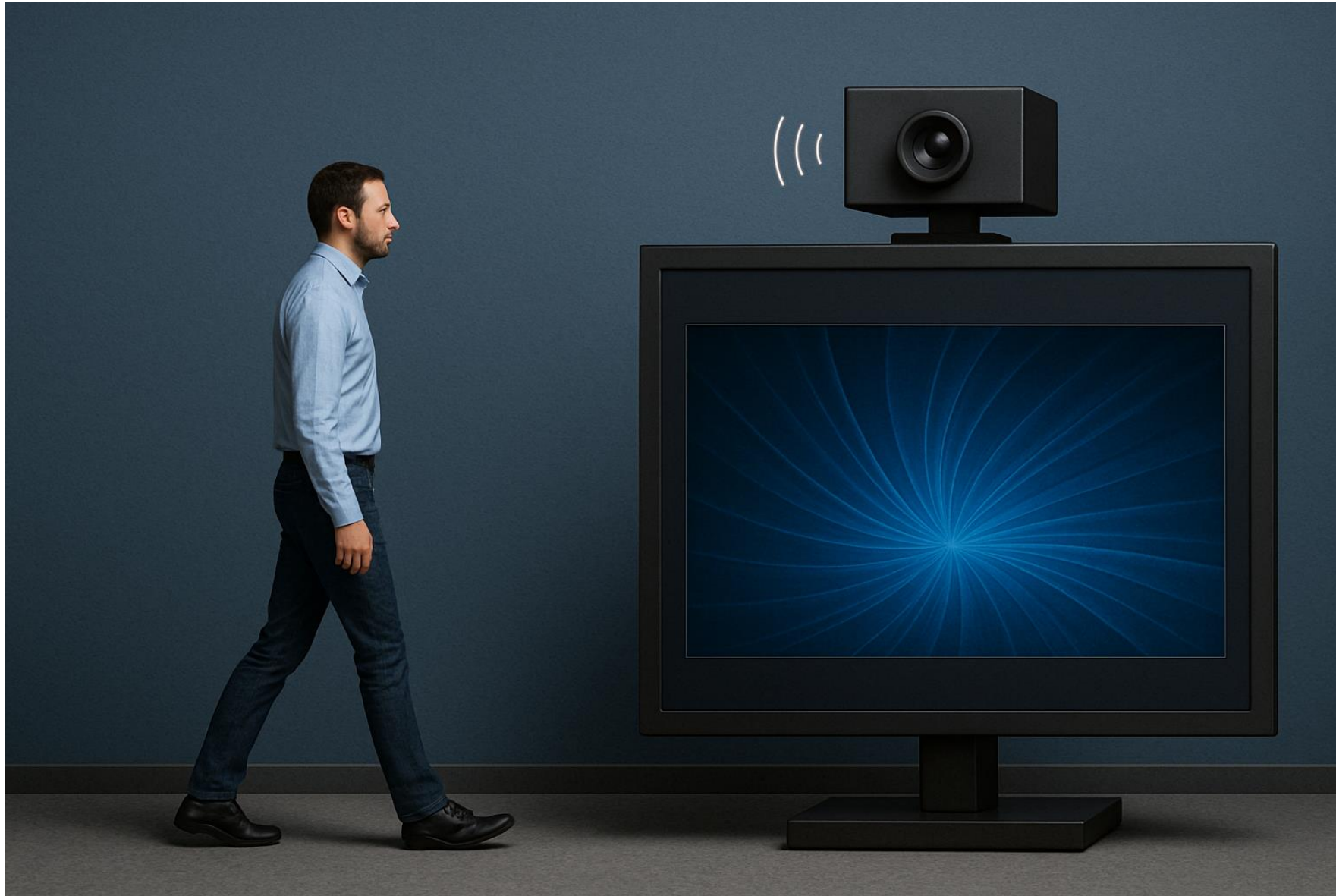


- Sensors
 - Sonar: SRF02
 - Camera: Sony PlayStation 3 Eye
- Processors
 - Microcontroller: Arduino uno for sonar sensor
 - Microprocessor: Raspberry pi 4 for camera and monitor
 - Communicating using PWM pins
- Operating system: Raspberry Pi OS (Raspbian)



- Streamed video is based on...
 - Face detection model (When sonar has sent success signals):
 - based on typical human facial features
 - for the mere detection of human presence
 - Age detection model (Only in Face detection state):
 - Kid
 - Adult
 - old
 - Gender detection model (Only in Face detection state):
 - Male
 - female
 - LBF model (Only in Face detection state):
 - facial landmark detection algorithm
 - estimates the positions of key facial points such as corners of eyes, mouth, nose tip, etc
 - Other facial features (Only in Face detection state):
 - Skin color, Hair color, etc
- Streamed audio is based on...
 - Face detection model (When sonar in not sending successful signals):
 - based on typical human facial features
 - Detecting faces at distance
 - Clothing Color Recognition

Playing audio based on simple face detection and clothing color recognition



Subject is intrigued, meanwhile more advanced models are processing captured video

